

Now I will explain **Unit-2: Cloud Computing** from your PDF in **very simple, easy and exam-friendly language**.

I will give:

- ✓ Points (for short answers)
 - ✓ Paragraph explanation (for 10–15 marks)
 - ✓ Examples (real companies)
 - ✓ Clear structured explanation of EACH topic
-

✓ 1 Web-Based Business Services

✦ Meaning

Web-based business services are **online services provided through websites or web applications**.

They are:

- Hosted on cloud servers
 - Accessed through internet
 - Used by businesses
-

✓ Examples from Your PDF

1. Cloud Storage & File Sharing

Examples:

- Google Drive
- Dropbox
- Microsoft OneDrive

Used for:

- Storing files online
- Sharing documents
- Collaboration

2. Project Management Tools

Examples:

- Trello
- Asana

Used for:

- Assigning tasks
 - Tracking progress
 - Team collaboration
-

3. CRM (Customer Relationship Management)

Examples:

- Salesforce
- Zoho Corporation

Used for:

- Managing customer data
 - Tracking sales
 - Automating marketing
-

4. Email Marketing

Examples:

- Mailchimp

Used for:

- Sending bulk emails
 - Marketing campaigns
-

5. Online Payment Services

Examples:

- PayPal
- Stripe

Used for:

- Accepting online payments securely
-

✓ Paragraph Explanation

Web-based business services are cloud-based applications that help businesses perform daily operations like storage, marketing, HR, accounting, and payment processing. These services are accessed through browsers and do not require installation. They reduce infrastructure cost and improve flexibility.

✓ 2 ☐ Delivering Business Processes from the Cloud

✦ Meaning

Running company processes (HR, Sales, Projects) using cloud software instead of local systems.

✓ Examples

- CRM via Salesforce
 - HR software in cloud
 - Project management via Asana
-

✓ Benefits

- Remote access
- Scalability

- Automatic updates
 - Lower maintenance
-

✓ Paragraph

Delivering business processes from the cloud means organizations use cloud-based applications to run operations like payroll, customer service, and project management. It improves efficiency and reduces hardware dependency.

✓ 3 ☐ Broad Approaches to Cloud Migration

There are 4 main approaches:

1 ☐ Lift and Shift (Rehosting)

✓ Meaning

Move application to cloud without changing code.

✓ Example

Moving local app to AWS EC2.

✓ Features

- Fast migration
 - Low cost
 - No code change
-

2 ☐ Replatform (Refactor)

✓ Meaning

Modify application to use PaaS services.

✓ Benefits

- Better scalability
 - More cloud advantages
-

3 ☐ Rearchitect (Rebuild)

✓ Meaning

Completely rebuild application using cloud-native technology.

✓ Benefits

- Maximum scalability
 - Modern architecture
 - Future-proof system
-

4 ☐ Data Migration

Move databases to cloud storage like:

- AWS S3
 - Google BigQuery
-

✓ Paragraph

Organizations can migrate to cloud using different strategies depending on cost, time and business goals. Lift and shift is easiest, while rearchitecting gives maximum benefits but requires more time and effort.

✓ 4 ☐ Seven-Step Model of Cloud Migration

Step 1: Assessment & Planning

Choose correct provider.

Step 2: Isolation

Prepare and clean data.

Step 3: Mapping

Choose cloud storage.

Step 4: Design & Architecture

Set up virtual machines and deploy apps.

Step 5: Augmentation

Use advanced cloud features.

Step 6: Validation & Testing

Test system performance.

Step 7: Monitoring & Maintenance

Regular updates and monitoring.

✓ Paragraph

The seven-step migration model ensures smooth transition to cloud. It includes planning, data preparation, deployment, testing, and continuous monitoring for long-term success.

✓ 5 ☐ Efficient Steps for Cloud Migration

Important Steps:

1. Assess infrastructure
2. Choose provider
3. Select migration strategy
4. Plan migration

5. Migrate data
 6. Migrate applications
 7. Optimize
 8. Security & compliance
 9. Training
 10. Continuous monitoring
-

✓ Paragraph

Efficient cloud migration requires proper planning, security implementation, and continuous monitoring. Organizations must train employees and follow best practices to maximize benefits.

✓ 6 Measuring and Assessment of Risks

✦ Meaning

Identify and measure cloud security risks.

✓ Steps

1. Identify risks
 2. Evaluate assets
 3. Threat analysis
 4. Vulnerability assessment
 5. Risk measurement
 6. Compliance check
 7. CSP evaluation
 8. Third-party risk
 9. Risk mitigation
 10. Continuous monitoring
-

✓ Paragraph

Risk assessment in cloud computing involves identifying threats, analyzing vulnerabilities, and implementing security measures to protect data confidentiality, integrity, and availability.

✓ 7 ☐ Company Concerns & Risk Mitigation

Common Concerns

1. Data security
 2. Downtime
 3. Compliance
 4. Vendor lock-in
 5. Data ownership
-

Risk Mitigation Methods

- Encryption
 - Access control
 - Multi-cloud strategy
 - Disaster recovery
 - Compliance audits
-

✓ Paragraph

Organizations adopt encryption, access control, and multi-cloud strategies to reduce cloud risks. Regular audits and compliance checks help maintain secure cloud environments.

✓ 8 ☐ Case Studies (Very Important for Exam)

1 ☐ Netflix

Concern: Downtime
Solution: Multi-cloud strategy
Result: High availability

2 Capital One

Concern: Data security
Solution: Encryption & audits
Result: Compliance maintained

3 Zoom Video Communications

Concern: High demand during COVID
Solution: Scaled using AWS & Oracle
Result: Reliable service

4 Zomato

Concern: Peak traffic
Solution: Dynamic scaling
Result: Smooth service

5 ICICI Bank

Concern: Vendor lock-in
Solution: Multi-cloud
Result: Flexibility

Final Unit–2 Summary

Unit–2 focuses on:

- Web-based business services

- Cloud migration strategies
- Risk assessment
- Security concerns
- Real company case studies

Cloud computing helps businesses reduce cost, improve scalability and increase flexibility. However, proper risk management and migration planning are necessary.

If you want next:

- 📄 10–15 mark answers ready to write
- 📝 Short notes for 5 marks
- 🎯 Important exam questions
- 📊 Comparison tables
- 📖 Quick revision sheet

Tell me